Chapter 2: The Linear Resonance Spectrum and Harmonic Particle Mapping

2.1 The Unified Frequency Foundation

The BMI framework resolves the diversity of the subatomic catalog into a single, cohesive linear resonance spectrum. Rather than treating leptogenesis and hadronization as independent phenomena, the BMI model demonstrates that the Electron, Muon, Pion, and Nucleons are fundamentally linked as integer harmonic modes of a unified base frequency, ω_0, modulated by the macroscopic evolution of **Bridge Stress (τ)** and **Inter-brane Potential (V_{gap})**.

2.2 Quantization of the Spectral Line

The mapping requires identifying integer harmonic indices (n) where the bimodal tension reaches a phase-locked state. The effective rest mass M_n is subject to **Thermodynamic Scaling**: it is a function of the harmonic mode, the base frequency, and the local metric equilibrium (Δ).

$$M_n(t) = n \cdot \omega_0 \cdot (1 + \Delta\psi(n)) \cdot f(\Delta(t))$$

This links the microscopic mass of particles directly to the evolution of the manifold coalescence ($\Delta = \tau + V_{gap}$) of the universe at time t.

2.3 The Core Particle Harmonic Catalog

- **The Leptonic Foundation:** The electron is the primary manifestation threshold ($n = n_{base}$), representing the baseline interaction between σ_A and σ_B.
- **The Muon Resonance:** A meta-stable excitation ($n = n_\mu$) maintained by local Bridge Stress (τ) that relaxes as the manifold settles.
- **The Mesonic Bridge:** A transition state ($n = n_\pi$) linking leptonic geometries to complex baryonic systems.

- **The Baryonic Cluster:** Complex nested phase-locks ($n = n_p, n_n$) where the global manifold geometry stabilizes.

2.4 Analytical Sweep, Impedance, and Mass Gaps

The high linearity of the spectrum is governed by **Geometric Impedance**. The mass gaps are induced by the Inter-brane Potential (V_{gap}). As the universe evolves toward **Asymptotic Flatness**, the Bridge Stress (τ) driving these higher-order modes diminishes, implying that in the ultra-distant future, the "particle zoo" will simplify as higher-order excitations become energetically impossible, leaving only base-mode leptonic configurations as the universe settles into its final, perfectly aligned geometric state.